

essay here of the value of producing new narratives that would make Islamicate history of science legible to colleagues specializing in other parts of the world. One way of accomplishing this goal, according to Stearns, is to produce histories that are regionally, temporally, or thematically anchored. He argues that studying the history of natural sciences in the Islamicate world needs to be informed by political, social, and intellectual contexts. As such, local or regional histories can help develop new insights that would help connect the scholarship on the Islamicate world to larger historical narratives.

Despite the perennial problems of access to sources (i.e., most of source materials in Islamicate science and medicine are unpublished, unedited manuscripts that are often poorly cataloged) and resources (i.e., support for the history of the Islamicate world is slim, not independent from the shrinking funding for the humanities in general), the study of Islamicate history is not only essential to the field of history of science for producing new knowledge, but also indispensable for the critique and correction it can offer to arguments drawn exclusively from the European experience and has thus the potential to set the agenda for future directions in history of science. The historiography of the Black Death is one such example among many that throws into sharp relief the significance of the Islamicate world.

“An Islamic Scientific Revolution? Early modern occult science, cosmic philology and the weird”

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Although I am an Islamicist, my project, focused on the early modern Persianate history of the occult sciences, is inescapably tied to the European Renaissance-then-Scientific Revolution narrative. Rather than dismiss it as terminally teleological, parochial, and Eurocentric, or simply irrelevant to the Islamicate context, I seek to reconceptualize this narrative as a challenge to Islamicists and Europeanists alike. As currently thematically and institutionally structured, the history of Western science as a field continues to pivot on two colonialist-Orientalist axioms:

- 1) Islam, eternally medieval, can only ever be an influence on the West, never the West itself.
- 2) Islamic occult science, unlike Christian, cannot even be protoscience, only pseudoscience, though it may magically be made good religion through conflation with “mysticism.”¹¹⁹

119. The Eurocentrism endemic to its modern study is a product of Renaissance humanist fictions, which went so far as to transform indispensable medieval Muslim authorities like Ibn Sina and Abu Ma'shar (Avicenna and Albumasar) into Europeans, while ignoring all contemporary sources. Arabic alchemy and medicine – the two disciplines are inseparable – are cases in point: some historians have presented them as protoscientific by stripping them of any *Islamicness*.

I propose that we retain the categories “Renaissance” and “Scientific Revolution” as heuristics, but greatly expand their intellectual-historical scope by way of the opposite, emically accurate axioms:

- 1) Islam is equally the West.¹²⁰
- 2) Occult science is simply science.¹²¹

Following our sources, the Islamicate occult sciences must no longer be defined according to the nineteenth-century colonialist triad *magic–religion–science*, but rather as mainstream methodologies designed to investigate and technologically harness mind–matter and mind–mind relationships, to extrapolate from visible to nonvisible data. Thus medicine was sometimes classified as an occult science, in contrast to surgery. Various modern disciplines like astrophysics and psychology are by this premodern definition often technically occult too – an epistemological continuity that should reassure the (we have never been) modern historian as to the relevance, the nonwretchedness, of her weird subject.¹²²

Due to the inherently Protestantizing – and hence *demonological* – nature of Islamic Studies from its inception, however, Islamicists tend to be more comfortable with the demonologists, rather than the mages, of Islam. Some historians have gone so far as to act as hammers of Muslim witches in their own right, burning their bodies and their books by deleting them from the narrative altogether. This is not historiography, but heresiography. Despite recent advances, the early modern Renaissance of Magic remains a major stumbling block to Islamicist historians of science.

A second stumbling block is the concept of Early Modernity itself. Recent research has shown that Muslim thinkers – especially occultists – were just as vociferous about their superiority over the ancients as their Christian peers, and just as insistent on the need for greater empiricism and experimentalism in the natural, mathematical, and metaphysical sciences. In astronomy-astrology, for example, it is this outlook that finally allowed the epochal break with Aristotelian physics. But such assertions of modernity and critiques of Aristotle began in the far more populous, wealthy, and cosmopolitan Islamicate world already in the thirteenth and fourteenth centuries, climate change, the Mongol conquest, and the Black Death being successive drivers, and reached a crescendo in the fifteenth and sixteenth, when a similar ethos finally began to pervade Latin Christendom as well. Why then has only the latter been awarded the descriptor “early modern”?

120. That is, that half of early modern Afro-Eurasia, encompassing half of the human race, where Hellenic-Abrahamic dialectics came to reign supreme.

121. My use of “occult science” and “Islamic occultism” is etic, yet tracks to emic categories too. Over the last millennium, Muslim encyclopedists regularly flagged the occult sciences (Ar. *al-‘ulūm al-gharība*) as a coherent, if shifting, subset of the natural and mathematical sciences, and increasingly the religious ones as well: for *gharīb* phenomena – or paranormal phenomena, in current English parlance – are common to all three categories of human experience.

122. I invoke here, of course, Neugebauer’s “The Study of Wretched Subjects” (note 58); and Latour’s *We Have Never Been Modern* (note 85).

A proof of divergence constantly invoked is the Latinate European embrace of printing, leading to a certain democratization of knowledge. But the Islamic world had already had its paper revolution many centuries earlier, which likewise led to a massive efflorescence of book culture, with private and public libraries orders of magnitude larger than those of contemporary Europe, circulating a much greater variety of titles.¹²³ Indeed, books were more socially important in Islamdom than in Christendom: its book culture fluidly substituted for more rigid Christian university and church hierarchies as primary scholarly framework. This efflorescence too entailed a certain democratization of knowledge, whereby books could now replace teachers, giving questions of scholarly authority and quality control a new salience.

Even more than their Christian counterparts, infinitely better studied, Muslim occultists were at the forefront of this cultural sea-change. From the fourteenth century onward, they began writing increasingly clear and comprehensive manuals on various sciences and establishing extensive, canonizing curricula. Why then has this vast body of explicitly scientific and technological literature been mostly excluded from our early modern historiography to date?

The remedy for this Christian-Islamic double standard, crassly colonialist, is a simple one, if strenuous: we must return to basic philology. The wild disproportionality in philological labor between Christian and Islamic sources may be epitomized here by a single case: geomancy. From the tenth century onward, this specifically Afro-Islamic divinatory discipline became ubiquitous throughout the Afro-Eurasian ecumene and thence the Western hemisphere, remaining popular globally to the present. In effect, it functioned as an early modern form of googling, surveilling, lie-detecting, and radar, among other state and personal uses, and is therefore integral to any history of technology; it even inspired Leibniz to invent binary computing.¹²⁴ The Arabic to Latin transmission of the earliest geomantic texts was garbled, truncated, and piecemeal, as with other translated sciences. Yet it enjoyed prestige as both “terrestrial astrology” and “celestial alchemy” throughout Europe too. A considerable body of scholarship has rightly been devoted to the roughly 150 Latin and vernacular geomantic manuscripts that survive in European collections.¹²⁵ But what of the tens of thousands of Arabic, Persian, and vernacular geomantic manuscripts preserved in archives internationally, to say nothing of the flood of lithographs and modern printings, almost all of which remain unstudied? And what of its many African and African-diasporic iterations globally?¹²⁶ Here as elsewhere, the proportion of philological labor to surviving archive is precisely reversed.

123. For examples see Matthew Melvin-Koushki, “*Tahqīq* vs. *Taqīd* in the Renaissances of Western Early Modernity,” *Philological Encounters* 3, nos. 1-2 (2018): 193–249.

124. Ron Eglash, *African Fractals: Modern Computing and Indigenous Design* (New Brunswick, NJ: Rutgers University Press, 1999), pp. 98–101.

125. The most comprehensive to date is Alessandro Palazzo and Irene Zavattero (eds.), *Geomancy and Other Forms of Divination* (Florence: SISMEL – Edizioni del Galluzzo, 2017), which includes three chapters on Hebrew and Judeo-Arabic forms of the science, but wholly excludes Muslim authors, giving the bizarre impression of the definitively Afro-Islamic science of geomancy as being somehow non-Islamic.

126. Ayodeji Ogunnaike, “The Dynamic Spread of Geomancy in Africa,” 19 October 2021, IOSOTR (islamicoccult.org).

Little wonder, then, that premodern (never modern) Islamic occultism has figured in the annual *Isis* bibliographies over the last century as a merest footnote, tantamount to African, Native American, and other “traditional” forms of *animism*. It has consistently been allotted half a page at most, and usually much less: of the fifty-five entries on Islam in the 2020 edition, 1 percent of the whole, only one relates, and only partially, to an occult science. This is in contrast to the dozens of pages of entries on Christian occultism, drowned in turn by the hundreds of pages of entries on nineteenth- and twentieth-century Euro-American science and technology, classed as neither Christian nor occult. Egregiously, the use of “pseudoscience” as a master category was only retired in 2002, when it was replaced by “occultism and natural magic” – but the underlying logic remains.¹²⁷ The same blatant teleology is institutionally reproduced throughout the discipline as a whole; bibliographic or geographic bias alone cannot explain such extreme disproportionality even on matters occult. While great strides have been made toward decolonizing the global history of science, exceptions continue to be implicitly and explicitly made for Islamic Magic as but a form of native animism, reprising a long tradition of militant Christian missionary arguments to the same end.¹²⁸

As an antidote, it is ethically imperative that we interrogate our own cosmologies, that we take more seriously the nonmaterialist panpsychist, idealist, and indeed animist cosmologies that boomed throughout the early modern West. It must be emphasized that our reflexive materialism – presupposing the irreality of mind, the basis of all occult science – is far from the only possible Western cosmology, a *final solution*. To the contrary, panpsychism remains a perfectly respectable option even in the Euro-American academy: the fey mysteries of quantum mechanics and the hard problem of consciousness have prompted its embrace by a small but prominent and growing cadre of scientists, philosophers, anthropologists, parapsychologists, and theologians working today. Some have even gone so far as to champion idealism as the ultimate scientific answer to materialism.

A similarly panpsychist cosmology impelled our early modern actors (including such Saints of Science as Bacon and Newton, and such Martyrs of Science as Bruno and Galileo) to embrace occult science as essential to their modernizing empiricism. It likewise drove a Renaissance of Pythagoreanism, whereby Muslims, Jews, and Christians strove together or in parallel to read, and magically rewrite, the cosmos as a second mathematical scripture, a project now hailed as the foundation of scientific modernity (the *mathesis* narrative). This Pythagorean paradigm shift – no less epochal than the Copernican, a subset of the same – I term *cosmic philology*.

Nor are such paradigm shifts of mere historical interest. In a late-capitalist era when the humanities are in existential crisis, and the natural sciences tied more to funding than to truth, we will need another Renaissance of philology if the academy is to survive

127. Shockingly antihistorically, “alchemy” is always neatly separated out from “chemistry” and “astrology” from “astronomy,” and both from “occultism.”

128. These arguments are epitomized by the American Protestant Reformed “Apostle to Islam” Samuel Zwemer’s (d. 1962) 1920 classic, *The Influence of Animism on Islam: An Account of Popular Superstitions*, which calls passionately for Islam’s destruction as a perverted, cancerous and above all Afro-Semitic tissue of demon worship and sorcery.

another plague- and conquest-ridden century. Deconstruction, while a necessary, quasi-kabbalist intervention appropriate to its late twentieth-century heyday, authorized the balkanization and decimation of the humanities over the last decades, and sent philology to the chopping block. The time has therefore come to reconstruct too. We must write better, fairer, more proportionate narratives of the past – and of the future.

I therefore propose that we begin to speak of an earlier Islamic “Occult-Scientific Revolution” in the same breath as the Christian “Scientific Revolution” itself, a category equally artificial yet equally heuristically useful. The question is whether it can eventually be integrated – or Integrated – into the standard history of science and technology curricula, given the enormous institutional inertia working against the field’s decolonization, as epitomized by its current textbooks.¹²⁹ But I am encouraged by the fact that my undergraduate students (I teach at a state flagship research university in the American South) seem to intuitively grasp the bizarre inequalities and fun-house distortions that define the history of science more than many specialists can, haunted as we are by the specter of the “superstition of the (Moslem) natives.” Our anxiety, however, is not unfounded: who knows what could happen if we take Islamic Magic as Western Science seriously?

That occult science specifically and Islamicate science generally are integral to any history of experimentalism and mathesis – and hence to the history of Western science – is corroborated in the contributions to this roundtable by Syed Nomanul Haq and Scott Trigg respectively. Our essays intersect fruitfully along a number of thematic axes, all of which must be pursued further. I will note a few strategic examples.

Trigg calls out the same colonialist-Orientalist classificatory system that still constrains the historical study of even “legitimate” Islamicate sciences like astronomy. Although by far the best-studied of such sciences to date, out of all proportion to the surviving archive, even this mathematical discipline has been caught in the trammels of the decline narrative, designed precisely to exclude Muslim intellection from the history of Western science to the greatest extent possible, to render it eternally medieval and Other. To counter this pernicious, antihistorical narrative, and drawing on a growing body of scholarship, he proposes two solutions: a return to philology, and an increased focus on mathesis.

129. Thoroughly representative is Andrew Ede and Lesley Cormack’s *A History of Science in Society: From Philosophy to Utility* (Toronto: University of Toronto Press, 2022), a widely used textbook now in its fourth edition. It explicitly and very briefly treats “classical” Islam as an appurtenance of Byzantium in chapter two, then spends the remaining eleven chapters charting the inexorable ascent to supremacy of Western European and then Euro-American “True Science”; there is not a single mention of most of the host of occult sciences, one mention of astrology and magic each as medieval irrelevancies, and only a few passages on alchemy emphasizing its death and replacement by chemistry. Even the most recent and otherwise excellent *Cambridge History of Science* volumes reflect this inertia: the volume on medieval science has several chapters on Islam but none on the occult, whereas the volume on early modern science has several chapters on the European occult but none on Islam. As a corrective, I develop the category of early modern Islamic Occult-Scientific Revolution in my chapter in Sonja Brentjes, Peter Barker and Rana Brentjes (eds.), *Routledge Handbook on the Sciences in Islamicate Societies* (note 19).

Philologically, we must reconceptualize the commentary genre as an often creative and quintessentially early modern one, not a form of sterile, eternally medieval scholasticism – for it is there that many of the signal contributions of Muslim scholars to “real science” were advanced.¹³⁰ This conceptual shift likewise allows intellectual historians to define a distinct Western early modernity: it is the period when Moderns began to respectfully disagree with the Hellenic Ancients. Only our membership in the nineteenth-century colonialist cult of origins can blind us to the creativity on display in “postclassical” Islamicate texts. What we deem modern science is not inherently more creative or original than these early modern investigations, as Kuhn, Latour, and other theorists have shown; but here again a Christian-Muslim double standard continues to be rigorously applied. To combat this historiographical connivance, Trigg holds up the paradigm-shifting mathematization of astronomy in the early modern Persianate world as an especially strategic node of comparative research going forward. I full-throatedly second both proposals, and argue only that our philological scope be extended to texts on associated occult sciences like astrology, lettrism (kabbalah), and geomancy – for they were equally important laboratories for mathesis and experimentalist creativity, in Islamdom and Christendom alike.

Alchemy was another such laboratory, as Haq has demonstrated. Classified by many of its Muslim practitioners as a simultaneously natural, mathematical, and metaphysical science, it embodied both mathesis and experimentalism, as Europeanists have been detailing for decades. Like many of its sister occult sciences, it served as a primary bridge joining non-Aristotelian (including Egyptian, Iranian, and Indic) scientific methods with Avicennan experimentalist natural philosophy, Ibn Sīnā’s own critique of the science notwithstanding.

But our rigid historiographical teleology has obscured this basic fact: Arabic alchemy, even more so than Latin alchemy, is generally seen as worth studying only to the extent it can be reclassified as “protochemistry” – and usually not even then.¹³¹ Haq rightly decries the resulting state of the field, whereby the greatest post-Jabirian Muslim alchemists remain almost entirely unstudied. They include in the first place the Cairene Aydemir al-Jildakī (fl. mid-fourteenth c.), whose daunting oeuvre outstripped that of Paracelsus (d. 1541) in historical influence, particularly in the Ottoman context; but studies on the latter abound, while only a single forthcoming monograph on the former has been produced to date, in the teeth of institutional inertia.¹³² This exclusionary,

130. Such a reconceptualization has been underway for the last few decades, though is much more advanced among Europeanists than Islamicists, true to the general historiographical pattern described here.

131. Here “Arabic” and “Latin” refer only to the language of composition of relevant works, not to Islamicate or Christianate scholarly cultures more broadly. Persian alchemical texts remain wholly unstudied, to say nothing of those in other Islamicate languages. By contrast, Greek alchemy has begun to receive scholarly attention; see Matteo Martelli, “Graeco-Egyptian and Byzantine Alchemy,” in Georgia L. Irby (ed.), *A Companion to Science, Medicine and Technology in Ancient Greece and Rome*. (London: Wiley-Blackwell, 2016), vol. 1, pp. 217–31.

132. Nicholas Harris, “Better Religion through Chemistry: Aydemir al-Jildakī and Alchemy under the Mamluks” (PhD dissertation, University of Pennsylvania, forthcoming).

teleological, Eurocentric, colonialist pattern has long constrained the study of the Islamicate occult sciences as a whole.

A recent surge in scholarship on the subject has broken this pattern, however, putting forward new taxonomies of knowledge that are philologically and codicologically faithful to our medieval and early modern sources, rather than to modern scientific, presentist chauvinism.¹³³ By a stroke of fortune, those taxonomies turn out to tally fairly well with the surviving and vast archival record, despite colonialist, reformist, nationalist, and Salafist depredations of the last two centuries. In the major collections of Iran, India, Egypt, and Turkey, for example, the natural and mathematical sciences – aka “real sciences” – appear to account for roughly twenty percent of total Islamicate scholarly production, across many millions of texts preserved in manuscript and lithograph form, which percentage accords with all the major premodern Arabic and Persian classifications of the sciences. And of that percentage, roughly *half* of these sciences are formally classified as occult.

To dismiss Islamic occultism as irrelevant Eastern, Southern, and Native mysticism is therefore to leave at least half of the history of Islamic Science unwritten. Such an act of unwriting, in turn, makes impossible an accurate, proportionate, decolonized history of Western Science as it was actually thought and practiced by Muslims, Jews, and Christians, separately and in tandem, over the last millennium. Past and present, the human experience of the world is often *weird*.¹³⁴

“Despair or opportunity? Coming to terms with the diversity of alchemy within its historical contexts”

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Thanks to some leading historians of medieval and early modern European science, European alchemy is now beginning to figure on the landscape of the history of science after its protracted banishment. While the historiography of alchemy has still not quite ceased to generate a good deal of controversy, we have indeed come a long way from professional historians’ summary dismissal of this art (*fann*, as the Arabic alchemist Jābir ibn Ḥayyān calls it) as “a pathology of thought,” “lunacy,” or an “obstacle” to the development of science.¹³⁵ And yet, the favorable scholarly push of

133. For the newest research see IOSOTR (islamicoccult.org).

134. In this Arabo-Persian context, “weird” translates *gharīb* and its semantic range more accurately than “occult,” signifying, in its Old English form *wyrd*, those uncanny, often traumatic workings of fate and nature that impel *wonder*. Compare here the now stock scientific term *quantum weirdness*. See Travis Zadeh, *Wonders and Rarities: The Marvelous Book That Traveled the World and Mapped the Cosmos* (Cambridge, MA: Harvard University Press, 2023); Erik Davis, *High Weirdness: Drugs, Esoterica and Visionary Experience in the Seventies* (Cambridge, MA: MIT Press, 2019), pp. 11–14.

135. This dismissive thrust of the eighteenth and nineteenth centuries is fully discussed in Lawrence Principe, “Alchemy Restored,” *Isis* 102 (2011): 305–12.